

and an 8-bit digital-to-analogue conversion (DAC) chip. As shown in the circuit diagram, you may also replace ATmega328 with the Arduino Uno board directly by connecting SCL and

SDA pins of Arduino to PCF8591.

In this circuit, the I2C bus connections include pins 9 and 10 of PCF8591 and pins 27 and 28 of ATmega328, which are SDA and SCL pins, respectively.

In this I2C bus, we can connect eight slave ICs (PCF8591) with each slave having a unique address for data transmission and reception. However, only five slave ICs are used in this project.

Table 1 shows the slave PCF8591 addressing in detail. A2, A1, and A0 are slave address bits to configure I2C slave address selection for different inputs (refer Table 2). To enable each slave IC, the corresponding address bits must be either GND (logic 0) or connected with 5V supply (logic 1) as shown in Fig. 1.

To know more about the slave address bits selection, refer to the control byte of PCF8591 as shown

Slave Address for Different Inputs

Table 2

A2	A1	A0	Analogue input
0	0	0	IC1 (port1)
0	0	1	IC2 (port2)
0	1	0	IC3 (port)
0	1	1	IC4 (port4)
1	0	0	IC5 (port5)

PARTS LIST

Semiconductors:

IC1-IC5 - PCF8591 8-bit ADC and DAC converter

IC6 - ATmega328 MCU

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R20, R22 - 10-kilo-ohm

R21 - 330-ohm

VR1 - 10-kilo-ohm potmeter

Capacitors:

C2, C2 - 22pF ceramic disk

Miscellaneous:

LCD1 - 16 $\times$ 2 LCD

Xtal1 - 16MHz crystal

CON1 - 2-pin terminal connector

CON2 - 2-pin connector

- 20-pin male Bergstrip for I/P supply

- 28-pin IC socket for IC6

- 16-pin IC socket for IC1 through IC5

- 16-pin female Bergstrip for LCD

- 20-pin male Bergstrip

(All these are available from kitsnspares.com)

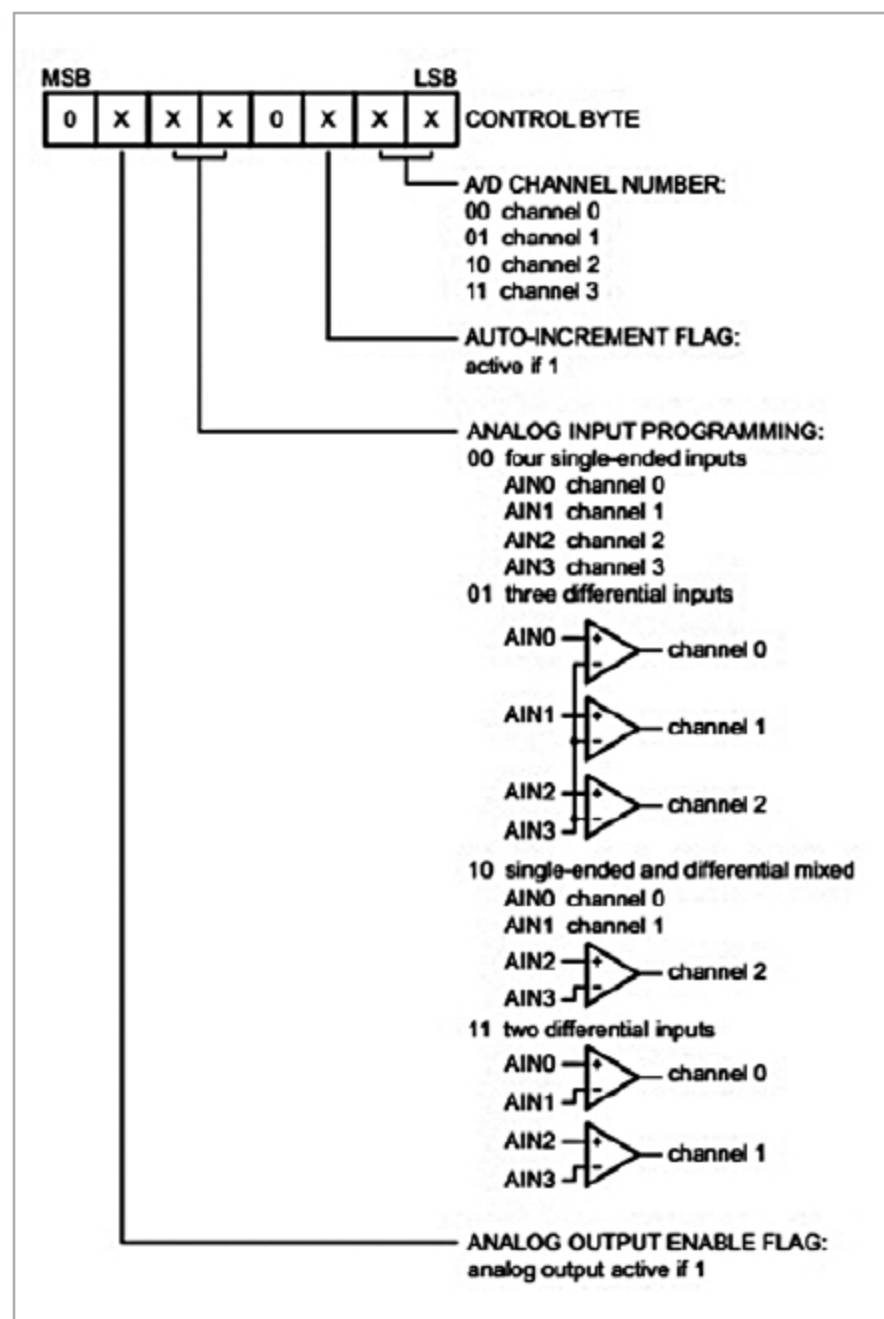


Fig. 2: PCF8591 I2C control byte

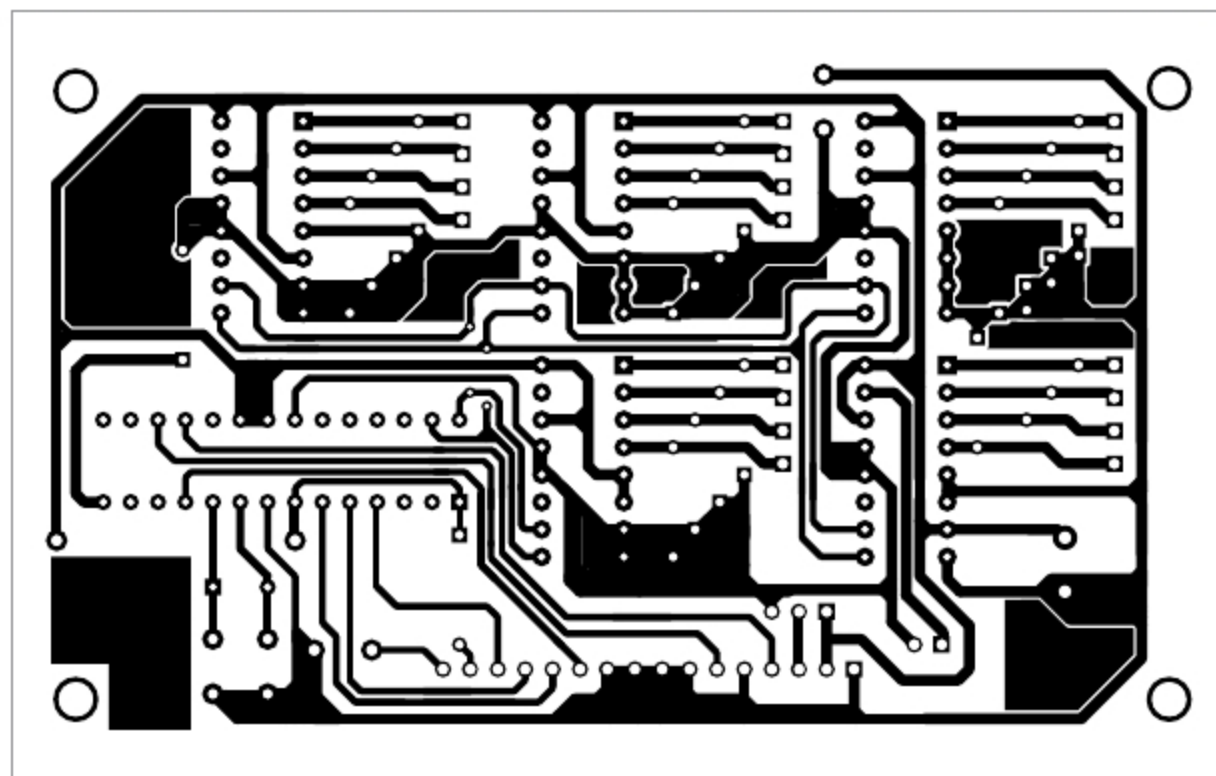


Fig. 3: Actual-size PCB layout for the port expander

in Fig. 2. By enabling the control bits we can configure the analogue inputs and outputs. Each slave IC has four analogue inputs (AIN0 through AIN3). In this project, there are five slave ICs, which means twenty analogue inputs (I/P1 through I/P20) of PCF8591 will be automatically scanned and the corresponding outputs can be displayed on 16 $\times$ 2 LCD display.

To get the outputs on the LCD, control byte and analogue output values are to be programmed in the coding. The system scans the addresses (A2, A1 and A0) of each slave port one by one, processes the inputs (AIN0 through AIN3) and displays them on the LCD. For better accuracy in the display, each analogue input pin is pulled down to ground using 10-kilo-ohm resistor.

Software

Arduino IDE is used for programming the code (Atmega328_expander.ino).